



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

IMPORTANT: This course is for the Double Major pathway only.

Programme	B.Sc. Physics Honours				
Course Title	RENEWABLE ENERGY SOURCES				
Type of Course	Value-Added Course 1				
Semester	III				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	0	45
Pre-requisites	Basic knowledge of different forms of energy.				
Course Summary	This course provides a comprehensive introduction to various renewable energy resources with a focus on non-conventional sources. Students will explore the principles, technologies, advantages, disadvantages, and practical applications of solar, wind, geothermal, ocean, and biomass energy.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Develop a foundational understanding of energy resources, focusing on non-conventional sources such as solar energy, and grasp key terms and concepts including solar constant, radiation measurements, collectors, and practical applications of solar power.	U	C	Instructor-created exams / Quiz
CO2	Discover wind energy comprehensively, covering utilization, advantages, disadvantages, environmental impact, sources, conversion principles, components, pros and cons, wind-electric power plants, economics, and operational challenges of large generators.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Gain insight into geothermal energy, exploring Earth's interior structure, geothermal systems like hot springs and various resources, and understanding the advantages, disadvantages, and applications of geothermal energy in comparison to other forms.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Explore ocean energy, focusing on tidal and wave energy, understanding tidal power plant components, economic aspects, OTEC working principles, efficiency, types, and applications, considering advantages and disadvantages.	U	C	Instructor-created exams / Home Assignments

CO5	Understand biomass with its resources and conversion processes, explore biogas applications and plants	Ap	P	Writing assignments
CO6	Study fuel cells, hydrogen energy, government schemes, and subsidies, and conduct plant visits for performance analysis.	Ap	P	Seminar Presentation /Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (36 +9)	Marks (50)
I		Solar Energy	12	18
	1	Introduction to Energy Resources-NonConventional Energy Sources-Renewable and Non-Renewable energy sources.	2	
	2	Solar Energy Terms and Definitions: Solar radiation, Solar Constant	2	
	3	Measurement of Solar radiation, Solar energy collectors: Principle (Physical) of the conversion of solar energy into heat, Types of collectors,	3	
	4	Flat plate collector, Advantages, disadvantages and applications of flat plate collectors,Need of orientation in concentrating collectors, Advantages and disadvantages of concentrating collectors, Parabolic trough collector	3	
	5	Concentrating collectors (Performance analysis not needed) ,Solar air heaters and dryers,	1	

	6	solar cookers, Solar photovoltaic cells (no need of mathematical equations)	1	
	Sections 1.3, 1.4, 1.5, 2.2.1, 2.2.2, 2.3, 3.1.3, 3.2, 3.3.1, 3.3.3, 3.4 - (excluding 3.4.11), 4.16, 4.17, 4.21.4 of Book 1			
II	Wind Energy		7	10
	7	Introduction, Utilisation aspects of wind energy, Advantages and Disadvantages of wind energy, Environmental impact of wind energy	1	
	8	Sources/Origins of wind, Principle of wind energy conversion and wind power, Pattern factor	2	
	9	Basic components of wind energy conversion system(WECS), Advantages and Disadvantages of WECS	2	
	10	Wind-Electric Generating Power Plant, Problems in operating large wind power generators.	2	
	Sections 5.1-5.6, 5.8- 5.10, 5.11, 5.20, 5.26 of Book 1			
III	Geo Thermal Energy		8	12
	11	Introduction to Geothermal energy, Important aspects of Geothermal Energy, Structure of Earth's interior, Geothermal system-Hot Spring structure.	2	
	12	Geothermal Resources -Hydrothermal, Geopressured	3	
	13	Geothermal Resources - Petro-thermal system, Magma Resources	2	
	14	Advantages and disadvantages of geothermal energy over other energy forms, application of geothermal energy	1	
	Sections 7.1, 7.2, 7.3, 7.5, 7.8.1, 7.8.2, 7.8.3, 7.8.4, 7.9, 7.10 of Book 1			
IV	Energy from Ocean		9	10
	15	Ocean Energy, Ocean Energy Sources, Tidal energy	2	

	16	Components of a Tidal Power Plant, Advantages and disadvantages of tidal power, Economic aspects of tidal energy conversion	2	
	17	Wave energy, Advantages and disadvantages, Factors affecting Wave energy	2	
	18	Ocean Thermal Energy Conversion (OTEC), Working principle of OTEC, Efficiency of OTEC	2	
	19	Types of OTEC Plants (Closed system, Thermoelectric OTEC system), Advantages and Disadvantages and Applications of OTEC	1	
		Sections 8.1, 8.2, 8.3.1, 8.3.8, 8.3.14, 8.4.1, 8.4.2, 8.4.3, 8.5.1, 8.5.3, 8.5.4, 8.5.5.1, 8.5.5.5, 8.5.6 of Book 1		
V	Open Ended Module : Biomass and Chemical Energy		9	
	1	Biomass resources, conversion process and applications		
	2	Biogas applications and biogas plants		
	3	Fuel cells and Hydrogen energy		
	4	Government schemes and subsidies for renewable energy projects		
	5	Any renewable energy plant visit and performance analysis		

Books and References:

1. Non- Conventional Energy Sources and Utilisation by R.K.Rajput, S.Chand Publishers, 1st Edition (Book 1)
2. Nonconventional energy resources by G. D. Rai, Khanna publishers-2008 (Book 2)
3. Solar Energy by S. B. Sukhatme-Tata McGraw-Hill Publishing Company Ltd - 1997 (Book 3)

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	2	2	1	0	0	0	0	0	0	0	0	0
CO 2	2	2	2	2	1	0	0	0	0	0	0	0	0
CO 3	2	2	2	2	1	0	0	0	0	0	0	0	0

CO 4	2	2	2	1	0	0	0	0	0	0	0	0	0
CO 5	2	2	2	1	0	0	0	0	0	0	0	0	0
CO 6	2	2	1	2	2	1	0	0	0	0	0	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	